

TRACE: Memory in Motion

In-Context Learning with Recurrent Memory — Full Visual Essay

Artifact: Interactive Visual Essay • Dual-Engine Parity Verified (Python × TypeScript) • arXiv Grounded

The Central Falsifiable Claim: A fixed-size recurrent state can absorb an unbounded number of key–value demonstrations without growing in size, but when two demonstrations share overlapping (non-orthogonal) keys, the readout for one leaks the value of the other — a measurable, reproducible form of forgetting caused by fixed capacity, not a scripted failure. In our controlled experiment, increasing the state dimension allows construction of more mutually orthogonal keys and therefore reduces the interference we observe.

Explicit Non-Claim & Scope Boundary: TRACE is an educational toy model instantiating the additive special case named in BDH-CQ §3.2 (Engdahl et al., 2026, arXiv:2608.09888). It does **not** implement or reproduce proprietary BDH-CQ weights, deep non-linear update networks, or ~5% sparse non-negative synaptic activation vectors. TRACE uses an unconstrained dense linear matrix in $\mathbb{R}^{d \times d}$ to make the mathematical mechanics solvable by hand.

01 / The Hook: Lossless In-Context Memory

Imagine demonstrating three key–value associations to a recurrent model: red → 1.0, blue → 2.0, and green → 3.0. Instead of concatenating tokens into a growing list, the model updates a single, fixed-size 3×3 matrix.

When queried with green, the readout produces **3.0000** with **0.0000 error**. The state did not grow by a single byte, yet it retains all three facts with machine precision. This demonstrates the primary thesis: fixed physical memory does not imply inability to absorb sequential context.

02 / Two Ways to Remember Context

Standard Transformer (KV Cache)

- **Mechanism:** Concatenates key and value projections for every token into an append-only list.
- **Memory Footprint:** $O(T)$ — grows linearly with context length T .
- **Capacity:** Lossless key separation via softmax dot-product attention over all past positions.

Recurrent Synaptic Memory (TRACE / BDH-CQ)

- **Mechanism:** Ingests demonstrations directly into an evolving internal state tensor S_t .
- **Memory Footprint:** $O(1)$ — exactly d^2 scalars, constant for all T .
- **Capacity:** Geometrically bounded by state dimension d . When $T > d$, overlap is inevitable.

03 / The Computational Substrate

The educational substrate instantiates the additive linear attention update rule analyzed in BDH-CQ §3.2 Engdahl et al., 2026; Schlag et al., 2021 :

$$\begin{aligned} \text{State Initialization: } S_0 &= \mathbf{0}_{d \times d} \in \mathbb{R}^{d \times d} \\ \text{Demonstration Update: } S_t &= S_{t-1} + v_t k_t^T \\ \text{Normalized Readout: } \hat{y} &= (S_t \cdot k_q) / (k_q \cdot k_q) \end{aligned}$$

When querying for target key k_q after writing T demonstrations, the readout decomposes formally into:

$$\hat{y}_a = v_a + \sum_{b \neq a} [(k_a \cdot k_b) / (k_a \cdot k_a)] v_b$$

The cross-talk term **Interference**(k_a, k_b) = $(k_a \cdot k_b) / (k_a \cdot k_a)$ is derived directly from linear algebra. It is neither simulated nor randomized.

04 / Step-by-Step State Evolution (d=3)

Let $d = 3$, with carrier vector $c = [1, 1, 1]^T / \sqrt{3}$. Orthonormal keys k_1, k_2, k_3 are constructed via QR decomposition:

- **Step 0:** Initial state S_0 is all zeros (3×3).
- **Step 1:** Write red $\rightarrow 1.0$ via rank-1 outer product $v_1 k_1^T$.
- **Step 2:** Write blue $\rightarrow 2.0$ via $v_2 k_2^T$. State absorbs new synaptic patterns.
- **Step 3:** Write green $\rightarrow 3.0$ via $v_3 k_3^T$. Final state S_3 holds all three facts simultaneously.

Because $k_i \cdot k_j = 0$ for all $i \neq j$, the cross-talk sum vanishes completely, yielding exact zero-error recovery.

05 & 06 / Direct Manipulation & Geometric Capacity

In an unconstrained linear subspace $\mathbb{R}^d$, the maximum number of mutually orthogonal non-zero vectors is strictly d . This creates an exact capacity threshold:

- **Under-Capacity ($N \leq d$):** Gram-Schmidt QR decomposition allocates mutually orthogonal vectors. The reconstruction error is **MSE $\equiv 0.000$** down to floating-point precision (10^{-31}).
- **Over-Capacity ($N > d$):** Vectors are forced into linear dependence. Key vectors on the unit sphere S^{d-1} must have non-zero inner products, causing measurable value leakage.

07 / Interference from Overlapping Keys (d=2)

To observe interference in isolation, we demonstrate two non-orthogonal keys in $d=2$:

- Demonstration A: $k_A = [1.0, 0.0]^T$, value = **1.0**
- Demonstration B: $k_B = [0.6, 0.8]^T$, value = **9.0**

The dot product is $k_A \cdot k_B = 0.60$. The leak coefficient is exactly: **Interference**(k_A, k_B) = **0.60 / 1.0 = 0.60**.

Querying k_A after writing both facts yields: **$\hat{y}_A = 1.0 + 0.60 \times 9.0 = 6.4000$** . The readout matches the analytical formula to floating-point tolerance. This distinguishes genuine geometric capacity leakage from arbitrary corruption or same-key overwrites.

08 & 09 / Connections to BDH & BDH-CQ

The Dragon Hatchling [Kosowski et al., 2025 \(arXiv:2509.26507\)](#) proposes that attention can be understood as fast synaptic plasticity. TRACE illustrates this by showing that memory write operations are rank-1 updates to synaptic

weights.

BDH-CQ Engdahl et al., 2026 (arXiv:2608.09888) proves that recurrent state models can solve novel symbolic tasks without chain-of-thought tokens. TRACE isolates the exact additive special case (§3.2) to demonstrate how in-context demonstration learning functions mechanically inside a fixed matrix.

10 / Primary Literature Grounding & Verified Citations

1. **BDH-CQ (2026)**: Engdahl, Kosowski, Chorowski, Stamirowska, Bartoszkiewicz, & Uznański. *In-Context Learning in BDH Without Chain of Thought*, arXiv:2608.09888. Section 3.2 formalizes general recurrence $S_t = U_\theta(S_{t-1}, D_t)$ and the additive special case.
2. **The Dragon Hatchling (2025)**: Kosowski, Uznański, Chorowski, Stamirowska, & Bartoszkiewicz. *The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain*, arXiv:2509.26507. Foundations of biological fast synaptic state.
3. **Mamba (2023)**: Gu & Dao. *Mamba: Linear-Time Sequence Modeling with Selective State Spaces*, arXiv:2312.00752. Theoretical capacity analysis of fixed-size state spaces and sequential compression limits.
4. **Fast Weight Programmers (2021)**: Schlag, Irie, & Schmidhuber. *Linear Transformers Are Secretly Fast Weight Programmers*, ICML 2021. Proves formal equivalence between linear attention and rank-1 outer product memory updating.
5. **Linear Transformers (2020)**: Katharopoulos et al. *Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention*, ICML 2020. Constant $O(1)$ inference step complexity.